23 - S2 - Q3

Q :

	A. P.	A. N.
P. P.	126	11
P. N.	15	100

(a) accuracy , precision , recall and Fscall

(b) peaking

(c) 3 feature subset selection methods.

Solution (a) ① $accuracy = \frac{TP + TN}{TP + FP + FN + TN}$

$$\begin{array}{|c|c|} \hline 126 & 11 \\ \hline 15 & 100 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline TP & FP \\ \hline FN & TN \\ \hline \end{array}$$

$$= \frac{126 + 100}{126 + 11 + 15 + 100}$$

$$= \frac{226}{252}$$

$$= 89.6825\%$$

② $precision = \frac{TP}{TP + FP}$

$$= \frac{126}{126 + 11}$$

$$= 91.9708\%$$

③ $recall = \frac{TP}{TP + FN}$

$$= \frac{126}{126 + 15}$$

$$= 89.3617\%$$

④ $F\text{-}Score = \frac{2 \times precision \times recall}{precision + recall}$

$$= \frac{2 \times 91.9708\% \times 89.3617\%}{91.9708\% + 89.3617\%}$$

$$= 0.9065$$

(b) ① peaking phenomenon defination

For a finite N, by increasing the number of features, one obtains an initial improvement in performance, but after a critical value, further increase of the number of features resultes in an increase of the number of features results in an increase of the probability of error.

② Guide.

(1) In order to design a classifier with good generalization performance, the number of training samples, N, must be large enough

with respect to the number of feature (

(2) For a small number of training data, a small number of features must be used.

(3) If a large number of training date is available, a larger number of features can be used for better performance

(4) collect sufficient training data

(c) Feature subset selection algorithm

① the filter method

(1) the filter method doesn't include a classifier in the loop, instead, it employs class separability measures as the feature subset evaluation criterion

(2) Advantages: ⓐ This criterion doesn't require training of a large number of pattern classifiers and is therefore computationally more efficient

ⓑ The feature subset thus selected is independent of the final classifier used.
ⓒ As the evaluation is based on the data distribution rather than on the performance of a complex model, filter methods are less prone to overfitting

(3) disadvantages

ⓐ the assumptions made in estimating data distribution overlap may not match the true distribution well

ⓑ It might mean that the chosen feature are not optimal for the intended mode

② The wrapper method

(1) The wrapper method include a classifier in the loop and use classification performance such as accuracy or error rate as the feature subset evalution criterion

(2) Advantages

ⓐ optimized for specific classifier

(3) Disadvantages

ⓐ Evaluating many possible subsets by repeated training a classifier can be very time-consuming

ⓑ Risk of overfitting: Since the procedure optimizes the feature subset for a specific classifier on the training data.

③ The embedded method

(1) Built-in mechanisms to penalize the number of features in the model

(2) Advantages

ⓐ combine the qualities of filter and wrapper methods.

ⓑ Integrated and efficient

Because feature selection is conducted during model training, compred to wrapper methods, embedded methods often requiree less additional computation.

ⓒ balance betwen model performance and complexity

(3) disadvantages

ⓐ If a different clasifer is chosen, the selected features might not be optimal

ⓑ Interprreting the selection criterion or the impact of regularization parameters can be challenging

(c) charges in model or criteria require re-traing the model, so it is limited flexibility